Box Man: Static recompilation of Game Boy games with LLVM

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1 Introduction

Building software-based emulators has always been an interesting exercise for recreational and hobbyist programmers. There is something satisfying about running old, nostalgic games like The Legend of Zelda, Super Mario Bros., or Pokémon Red/Blue/Yellow through an emulator you built yourself.

Earlier this year, one of us got nerd-snipped into emulation development (commonly referred to as “emudev”) by this site called Visual 6502, which is a browser-based transistor-level 6502 emulator. From then on, we have built a couple of small-scale emulators for old 2D-based retro game systems like CHIP-8 and NES.

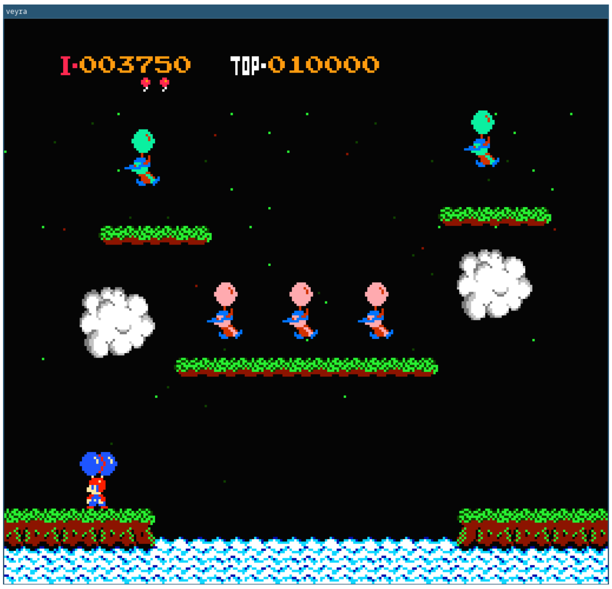

Figure 1: Balloon Fight running on tiny.nes

We wanted to build another emulator for the Game Boy, but this time we didn’t want to build just another generic interpreter. Around then, we came across a blog post by Andrew Kelley titled “Statically Recompiling NES Games into Native Executables with LLVM and Go”, and that’s where the idea for Box Man came from.

Box Man is a tool that takes in Game Boy ROMs and emits LLVM IR, which can then be compiled into a native executable using tools like `clang`. The resulting binary plays that specific game natively, without needing an external Game Boy emulator at runtime. It’s basically equivalent to ahead-of-time compilation of Game Boy ROMs.